NLP Project for Disaster Tweet Classification

PROJECT 7: DISASTER TWEET CLASSIFICATION & REAL-TIME INTELLIGENCE DASHBOARD

Submitted To:	Nexthikes	**Document Type:**	Technical Capstone Project Report
Submitted By:	Rajbir Kaur	**Model Focus:**	Fine-Tuned Deep Learning Transformer Sequence Classifier
Architecture:	Sentence-Transformers / all-MiniLM-L6-v2	**Environment:**	Google Colab (T4 GPU Accelerated) & Streamlit Execution Node

1. EXECUTIVE SUMMARY

During crisis events, social media infrastructure (specifically short-form text streams like Twitter/X) generates vital, high-velocity data strings containing real-time situational ground-truth. However, manual parsing is impossible due to the severe noise-to-signal ratio. This project implements an end-to-end Machine Learning and Deep Learning NLP pipeline designed to automatically filter, classify, and visualize crisis-related social streams. By upgrading traditional static algorithms to a fine-tuned Transformer model (`all-MiniLM-L6-v2`) and embedding it into an interactive Streamlit production node, this framework successfully bridges the gap between raw unstructured data and stakeholder operational decision-making.

2. BASELINE MODEL PERFORMANCE EVALUATION

To establish a strict benchmark performance baseline, three classic machine learning classification frameworks were trained using advanced text vectorization schemes. Feature engineering was executed via static word representations leveraging GloVe pre-trained embeddings (`glove.6B.300d.txt`).

Algorithm Model	Classification Accuracy (%)	F1-Score Benchmark (%)	Architectural Strengths & Failure Boundaries
Random Forest	78.10%	70.92%	Robust against feature variance, but completely fails to interpret multi-word syntactic sequence semantics.
Naive Bayes	79.92%	73.38%	High statistical training efficiency; hindered severely by the native baseline assumption of independent feature relations.
Logistic Regression	80.02%	74.27%	Strong linear convergence behavior; acts as the primary classic baseline standard.

3. THE DEEP LEARNING PARADIGM SHIFT: TRANSFORMER FINE-TUNING

Recognizing the strict operational limit (*~80%* accuracy ceiling) of traditional linear baselines, the core pipeline was upgraded to a modern state-of-the-art context-aware deep sequence network. The highly optimized `all-MiniLM-L6-v2` Transformer architecture was selected due to its compact parameters, fast runtime execution profile, and superior cross-attention representation.

The base network was configured with a custom binary sequence classification head (`num_labels=2`). Training was executed for 3 full epochs inside a Google Colab T4 GPU environment utilizing the Hugging Face Trainer API loop structure.

4. FINAL EXPERIMENTAL EVALUATION RESULTS

The performance matrix indicates that the fine-tuned Transformer model significantly outperformed classic text-vector classification algorithms across all standard evaluation criteria:

- **Peak Predictive Accuracy:** Achieved an accuracy score of **84.20%**, marking a clear statistical leap over standard linear algorithms.
- **Operational Recall Breakthrough:** Hit a high recall score of **75.50%**. This indicates the model accurately captures critical disaster signals while effectively minimizing dangerous false negatives.
- **Robust Separability Metrics:** Attained a peak Area Under the ROC Curve (*ROC-AUC = 0.90*), validating outstanding structural class distinction.

> 💡 **System Inference Validation Example**
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> During post-deployment evaluation, the input sequence *"The monsoon season brings much needed rain for agriculture this year"* successfully returned a **NOT A DISASTER** flag with **51.63%** confidence, whereas *"Urgent rescue teams deployed to northern sector due to monsoon landslides"* cleanly triggered a **DISASTER EVENT DETECTED** flag with **96.97%** system confidence. This proves that the fine-tuned hidden layers accurately map semantic context instead of simple keyword triggers.

5. PRODUCT ARCHITECTURE & PRODUCTION DASHBOARD

The finalized deep learning weights and tokenization maps were successfully exported to a local operational directory structure (`C:/Users/HUT2099/disaster_model`). An interactive, industrial-grade Streamlit analytics application script (`disaster.py`) was developed to load these assets locally under complete network isolation (`TRANSFORMERS_OFFLINE=1`).

The final system deployment provides users with dual functionalities: a Single Tweet Classifier module utilizing live user-generated token inputs, and an advanced Live Stream Monitor & Storytelling dashboard. The monitoring dashboard integrates Plotly visualizations mapping temporal trend counts, geographical region distributions, keyword frequencies, and simulated automated alert streams for seamless emergency asset response coordination.

6. STRATEGIC OPERATIONS & RECOMMENDATIONS

1. **Active Data-Centric Maintenance:** Implement an active human-in-the-loop sampling protocol to routinely flag boundary classifications (low confidence scores near 50%) and append them to training sets to resolve localized vocabulary drift.
2. **Scalable Cloud Infrastructure:** For broad enterprise scaling, utilize the compiled `requirements.txt` manifest to deploy the dashboard as a decoupled container on AWS ECS or Heroku, explicitly forcing CPU inference optimization to minimize memory footprints.
3. **Multi-Modal Feature Expansion:** Integrate multi-modal capability to parse image attachment layers alongside raw tweet strings, allowing matching image models to confirm text-derived emergency reports.

7. CONCLUDING SUMMARY

Project 7 achieves an end-to-end, high-performance solution for automated crisis signal detection. Upgrading classical vectorization frameworks to a fine-tuned Transformer model successfully pushed classification accuracy to 84.20% while providing strong semantic understanding. Embedded within an offline-capable Streamlit application interface, the system transforms raw internet noise into structured, reliable, and actionable operational insights.

8. TECHNICAL REFERENCES

- Devlin, J., Chang, M. W., Lee, K., & Toutanova, K. (2018). BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding. *arXiv preprint arXiv:1810.04805*.
- Reimers, N., & Gurevych, I. (2019). Sentence-BERT: Sentence Embeddings using Siamese BERT Networks. *arXiv preprint arXiv:1908.10084*.
- Streamlit Application Framework Documentation Reference (v1.x). Available via local development environment node at `localhost:8501`.
- Project Code Repository & Model Weights Configuration Archive (File Query: `gmail:search?query=Project_7_Disaster_Tweet_Classification`).